Boosting 2

Saturday, March 2, 2024 9:09 PM

- Mean

$$F(x)_0 = \underset{\gamma}{\operatorname{argmin}} \sum L(y - \hat{y})$$

Mean

$$\gamma_{im} = - \frac{\partial L(y, F(x)_0)}{F(x_0)}$$

After learning h1 (x) - We need to find γ

STEP 2 . 3

3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:

$$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$

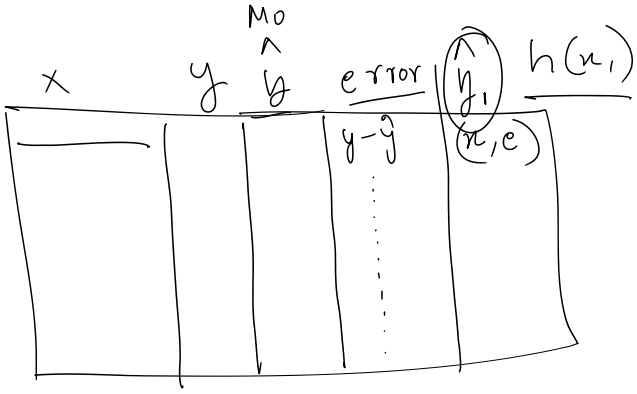


In simple terms, this equation is saying ;

- To find γ which minimizes given **loss**

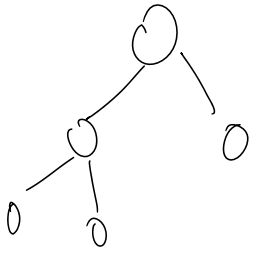
$$L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

y_i *Actual Value* — $F_{m-1}(x_i)$ $\hat{y}$ *(Predicted Value)*



$$F = \text{Avg} + \lambda \frac{\text{Newmodel}}{n(w)} + \text{error}$$

Stump



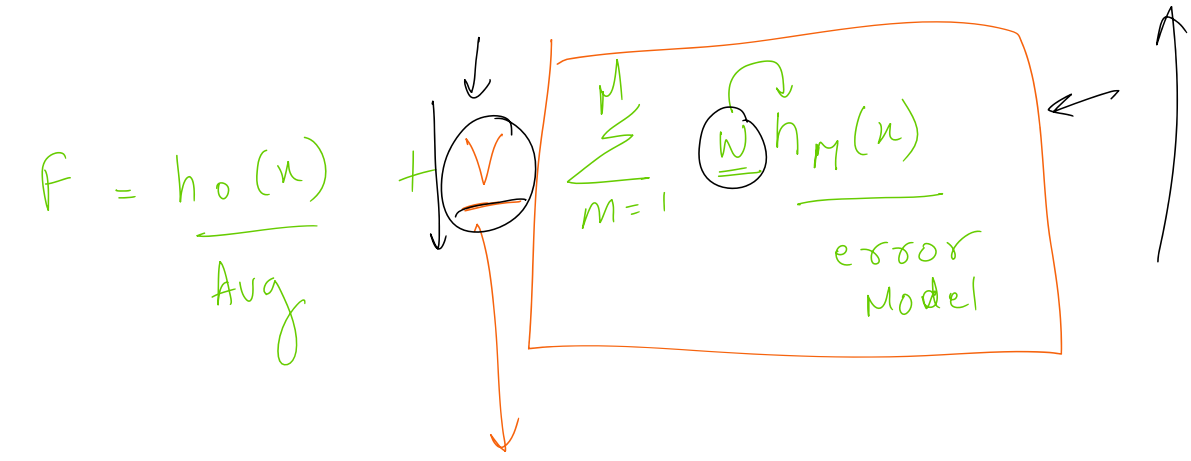
Hyper paramater

- weights ? ✗
- No of Model ✓
- Depth

$$y = w_1 x + w_2 x$$

Train

M_1 M_2 M_3 M_3 M_4 M_5 M_3 . . ↓ error



learning
regularization

66 + 15

66 + 1.5

so

0.001

66 15

$$F = h_0 x + \lambda h_1 x \quad \ldots\ldots$$

very small

1.5

GBDT

- overfitting

High variance

Regularization

RS + CS

Stochastic Gradient Boosting

